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Event Stream Filtering via Probability Flux Estimation

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Abstract

Event cameras asynchronously capture brightness changes with microsecond latency, offering exceptional temporal precision but suffering from severe noise and signal inconsistencies. Unlike conventional signals, events carry state information through polarities and process information through inter-event time intervals. However, existing event filters often ignore the latter, producing outputs that are sparser than the raw input and limiting the reconstruction of continuous irradiance dynamics. We propose the Event Density Flow Filter (EDFilter), a framework that models event generation as threshold-crossing probability fluxes arising from the stochastic diffusion of irradiance trajectories. EDFilter performs nonparametric, kernel-based estimation of probability flux and reconstructs the continuous event density flow using an O(1) recursive solver, enabling real-time processing. The Rotary Event Dataset (RED), featuring microsecond-resolution ground-truth irradiance flow under controlled illumination is also presented for event quality evaluation. Experiments demonstrate that EDFilter achieves high-fidelity, physically interpretable event denoising and motion reconstruction.

1. Introduction

Event cameras are bio-inspired vision sensors that asynchronously report brightness changes with microsecond latency [11, 22, 29]. Unlike conventional frame-based sensors that sample at uniform time intervals, the irregular, threshold-triggered event measurements avoid redundant sampling and thus fit into visual motion perception [9, 13, 14, 30]. But this new paradigm also magnifies the internal noise of the circuit, as pointed out in [4, 18].

By analyzing the event generation model [22], we observe that an event stream conveys two complementary types of information about the underlying brightness evolution, as illustrated in Fig. 1: (1) the **state information**, encoded by the discrete jump in logarithmic irradiance at the event time $I_{t_i} - I_{t_{i-1}} = p_i C$ and, (2) the **process information**, described by the inequality $\sup_{t\in[t_{i-1},t_i)} |I_t - I_{t_{i-1}}| < C$, which constrains the latent irradiance trajectory between successive events. Here I_t denotes the logarithm scene irradiance, $t_i \in \mathbb{R}$ and $p_i \in \{-1, 1\}$ the timestamp and polarity of the i-th event and C the contrast threshold. Classical signal processing approaches model continuous state variables effectively, but they are inherently unsuitable for discontinuous, asynchronous event polarities and cannot directly impose the above inequality constraint on the latent irradiance path. Consequently, most existing event filters ignore the process information and treat event polarities as if they were continuous signals, often producing outputs that are even sparser than the raw input and ultimately limiting the reconstruction of continuous irradiance dynamics.

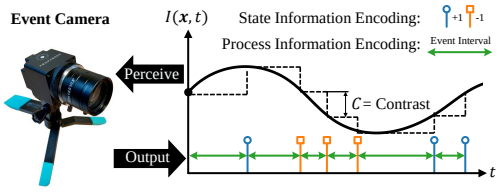
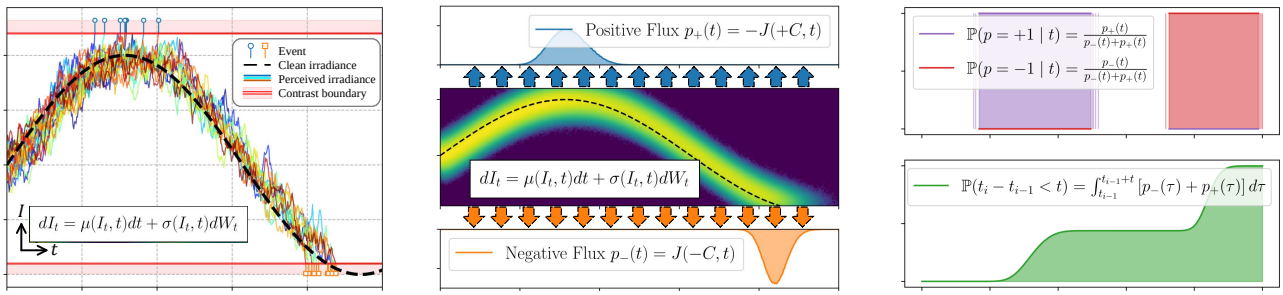


Figure 1. **Interpretation of the information encoded in an event stream.** Each event provides (1) state information through the discrete irradiance jump at the event time, and (2) process information by constraining the latent irradiance trajectory between events through the contrast-bound inequality.

To bridge this gap, we revisit event generation from a stochastic process perspective. Intuitively, an event is triggered whenever the observed brightness trajectory moves far enough to cross the ON/OFF threshold. This trajectory is affected by thermal noise and exhibits randomness over time. The likelihood of this crossing is not determined by the past events themselves but how the brightness is flowing toward the threshold at any moment. This directional tendency of irradiance trajectories to cross the boundary is precisely what the probability flux (or current) represents.

Formally, although events appear as discontinuities, the irradiance evolves continuously and can be modeled as a

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(a) A stochastic specification of irradiance evolution, where clean irradiance and thermal noise determine the drift and diffusion of observed irradiance trajectories, and an event is triggered at the contrast boundary.

(b) A distributional specification obtained by counting trajectory densities over irradiance–time space. The incoming probability flux at the contrast boundary corresponds to the instantaneous probability of generating an event.

(c) The distributions of discrete event polarities conditional on event time, and continuous inter-event times, obtained respectively from the ratio of polarity-specific boundary fluxes and their temporal integrals.

Figure 2. Stochastic analysis of event generation for a single pixel.

threshold-crossing stochastic process. We find that both (1) the continuous distribution of **inter-event intervals** $\mathbb{P}(t_i - t_{i-1} < t) = \mathbb{P}(\sup_{\tau \in [t_{i-1}, t_{i-1}+t)} |I_\tau - I_{t_{i-1}}| \geq C)$ and, (2) the discrete distribution of **event polarities** $\mathbb{P}(p_i = \pm 1 \mid t_i) = \mathbb{P}(I_{t_i} - I_{t_{i-1}} = \pm C \wedge \sup_{t \in [t_{i-1}, t_i)} |I_t - I_{t_i}| < C)$ are determined by the threshold-crossing probability fluxes, as shown in Fig. 2. Estimating this flux from discrete, noisy events provides a generative formulation of event filtering by resampling a clean, filtered event stream.

Building on this principle, we propose a generative event filtering framework termed Event Density Flow Filter (EDFilter) that estimates threshold-crossing probability fluxes in real time. The proposed EDFilter models the spatiotemporal density of events using a nonparametric kernel smoother optimized via recursive maximum likelihood estimation. This formulation leads to an $O(1)$ recursive solver, enabling online operation at sensor time scales.

We further propose the Rotary Event Dataset (RED), which provides microsecond-accurate ground-truth irradiance references under controlled motion and lighting conditions. Comprehensive experiments demonstrate that EDFilter achieves superior denoising accuracy and temporal fidelity compared with state-of-the-art methods, while maintaining theoretical interpretability and real-time efficiency.

In summary, the main contributions of this work are:

1. The interpretation of event sensing as a threshold-crossing stochastic process is proposed, revealing that both inter-event intervals and event polarities are determined by probability fluxes at the contrast boundaries.
2. A generative, probability-flux-based event filter EDFilter is proposed, which estimates boundary-crossing fluxes from discrete events using nonparametric, $O(1)$ recursive kernel smoothers, enabling real-time signal filtering.
3. The RED dataset with precise microsecond ground truth irradiance references is proposed, which enables precise evaluation of temporal fidelity and supports research on physically accurate event sensing models.
4. Extensive experiments on RED and existing datasets show that EDFilter achieves superior denoising accuracy and temporal precision.

2. Related Work

Discriminative Filters. Most filters are discriminative. Density-based methods [3, 10, 17, 20, 24, 33] pass only events with sufficient spatiotemporal neighbors. Motion-based methods [25, 31] fit local planes to estimate velocity and filter out events that are not motion-consistent. More recently, learning-based methods use MLP [16], CNN [1], or Transformer [19] on event patches to classify each event as signal or noise. The limitation of these methods is their discriminative nature: they can only remove events, not correct them or generate a new, clean stream.

Generative Filters. A few methods [5–7, 32] are generative, most notably those [6, 7] that convert events into stacked event frames and use a 3D-UNET to denoise and super-resolve these frames. While powerful, these methods sacrifice the core advantage of event cameras—temporal precision—by binning events into synchronous frames.

Our Approach. The work most closely related to ours is [23]. Although designed for event simulation, it models event generation through stochastic diffusion of irradiance for continuous-time event simulation. Our method takes the opposite direction: we infer the diffusion from the observed events. Crucially, we show that events can be interpreted as direct samples of boundary probability fluxes, allowing us to bypass solving the full diffusion equation and derive a principled and efficient formulation for event modeling.

3. Approach

3.1. Theoretical Foundation

This section derives the probability flux as a physical quantity for characterizing event distributions, as illustrated in Fig. 2. This is achieved by modeling the evolving irradiance trajectory using a stochastic differential equation

(SDE) with absorbing boundaries. For clarity, we focus on a single pixel and assume the process starts at 0, but the formulation naturally extends to multi-pixel settings with spatial coupling and interference effects. Detailed derivations are provided in the supplementary material.

For a given event pixel, let I_t denote the logarithmic irradiance intensity. Under realistic conditions—assumed to be Markovian and continuous—it follows the SDE:

$$dI_t = \mu(I_t, t)dt + \sigma(I_t, t)dW_t, \tag{1}$$

where the drift $\mu(I_t, t)$ captures clean scene dynamics and the diffusion $\sigma(I_t, t)$ models thermal noise, with W_t a standard Wiener process, as shown in Fig. 2(a). Each pixel contains ON and OFF comparators that continuously compare I_t against contrast thresholds $\pm C$. An event is triggered when I_t exits the interval $[-C, C]$, with **timestamp** $t_s = \inf\{\tau > 0 \mid I_\tau \notin [-C, C]\}$. The exit boundary determines the **polarity** $p \in \{+1, -1\}$.

Associated with this SDE is the Fokker-Planck equation [8], which states the probability density function $\rho(I, t)$ of I_t is the solution to the boundary value problem:

$$\partial_t \rho(I,t) = \frac{1}{2}\partial_I^2[\sigma^2(I,t)\rho(I,t)] - \partial_I[\mu(I,t)\rho(I,t)], \tag{2}$$

$$\text{subject to } \rho(-C,t) = \rho(C,t) = 0, \quad \rho(I,0) = \rho(I), \tag{3}$$

where $\rho(I)$ is the initial distribution. To simplify the expression, we introduce the probability flux density as

$$J(I,t) = \mu(I,t) - \frac{1}{2}\partial_I[\sigma^2(I,t)\rho(I,t)], \tag{4}$$

representing the instantaneous flow of probability mass, since it follows the continuity equation

$$\partial_t \rho(I,t) + \nabla \cdot J(I,t) = 0, \tag{5}$$

and the outgoing fluxes at the boundaries,

$$p_\pm(t) = \mp J(\pm C, t), \tag{6}$$

constitute the instantaneous ON/OFF event densities as shown in Fig. 2(b). Hence, events can be interpreted as samples drawn from the leaking probability flux at the boundary. Since the number of leaked events should be complementary to the remaining trajectories, we have

$$\begin{aligned} \mathbb{P}(t_s \le t) &= 1 - \int_{-C}^{C} \rho(I,t)dI, \\ &= \int_0^t [p_+(\tau) + p_-(\tau)]\, d\tau. \end{aligned} \tag{7}$$

The conditional distribution of polarities is derived as:

$$\mathbb{P}(p = \pm 1 \mid t) = \frac{p_\pm(t)}{\mathbb{P}(t_s = t)} = \frac{p_\pm(t)}{p_-(t) + p_+(t)}, \tag{8}$$

completing the proof of the results in Fig. 2(c).

However, the solution of Eqs. (2) and (3) does not generally admit a closed-form expression [8]. To address this, we reformulate them with the failure rate [12], here as the **event density flow (EDF)**:

$$\lambda_\pm(t) = \frac{p_\pm(t)}{1 - \int_0^t [p_+(\tau) + p_-(\tau)]d\tau}, \tag{9}$$

which describes the instantaneous expected event rate. The formulation of $\lambda_\pm(t)$ allows direct estimation from discrete events, since they are unconstrained nonnegative real-valued functions. In this work, we employ nonparametric temporal and spatial kernel-based smoothers to estimate these functions, as the following section shows.

3.2. The EDFilter Framework

The proposed EDFilter framework, illustrated in Fig. 3, consists of three sequential components that respectively predict, update, and reconstruct the event density flow.

3.2.1. Sequential Kernel Density Estimation

The first step is to generate a continuous density prediction at any given time. This is achieved using a temporal kernel density estimator (KDE) for each pixel independently. A rectangular kernel is adopted to model the contribution of sequential events, while both its height and bandwidth are treated as parameters to be estimated online. To account for noise, a constant false event rate is also introduced and optimized jointly. These defines our **temporal EDF model**:

$$(\lambda_+(t), \lambda_-(t)) = \begin{cases} (\psi(t), \beta_-), & \text{if } \psi(t) > 0 \\ (\beta_+, -\psi(t)), & \text{if } \psi(t) < 0 \\ (\beta_+, \beta_-), & \text{if } \psi(t) = 0 \end{cases} \tag{10}$$

$$\psi(t) = \sum_{t_i < t} p_i \phi(t - t_i), \phi(t) = \begin{cases} \alpha, & \text{if } 0 \le t < h \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where α and h are the height and bandwidth of the rectangular kernel $\phi(t)$, $\psi(t)$ is the predicted density, and $\beta_\pm \ge 0$ denote false event rates. Unlike existing event filters, abnormal events are not hard-thresholded but still contribute to the continuous scene modeling process.

To determine optimal EDF parameters, we adopt the likelihood maximization strategy. If N_e events are observed in the interval $(s, t]$ with times and polarities $\{(t_i, p_i)\}_{i=1}^{N_e}$, the log-likelihood is can be derived from Eqs. (7) and (8) as

$$\ln \mathcal{L}(s,t) = \sum_{i=1}^{N_e} \ln \lambda_{p_i}(t_i) - \int_s^t [\lambda_+(t) + \lambda_-(t)]dt, \tag{12}$$

which combines the contributions of state and process likelihoods, where $\lambda_p(t) = \lambda_\pm(t)$ for $p = \pm 1$. To derive closed-form solutions, we also impose a prior distribution:

$$f(\alpha, \beta_\pm, h) \propto \frac{\gamma}{h^2} \exp\left(-\frac{\gamma}{h}\right), \tag{13}$$

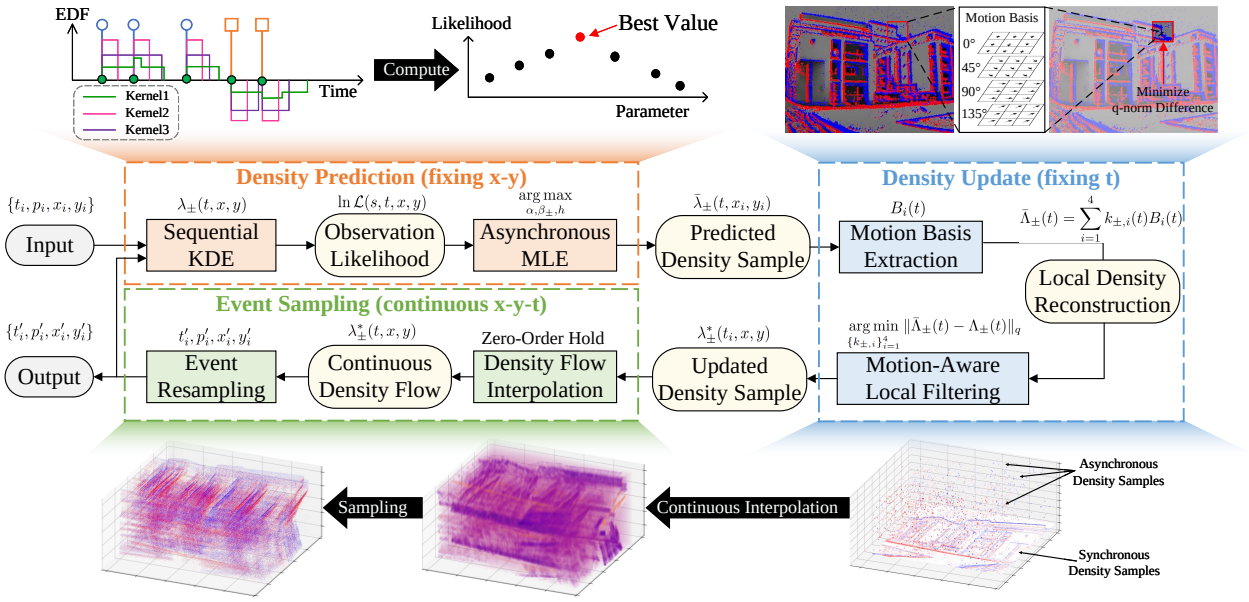


Figure 3. **The proposed EDFilter.** The density-prediction module sequentially applies KDE to the input events and selects the optimal kernel by maximizing the event-observation likelihood, producing a predicted density sample. The density-update module then spatially fuses these density samples using a motion-aware, sparsity-preserving local filter to obtain refined density estimates. Finally, the event-sampling module interpolates the continuous event-density flow via zero-order hold and resamples the filtered events for output. This output also triggers the density-prediction module in an application-dependent manner, mitigating the impact of abnormal integration.

where γ is interpreted as the expected event interval. Maximizing the posterior likelihood yields the predicted EDF:

$$\bar{\lambda}_{\pm}(t) = \lambda_{\pm}(t)|_{\arg\max_{(\alpha,\beta_{\pm},h)}[\ln \mathcal{L}(s,t)+\ln f(\alpha,\beta_{\pm},h)]}. \quad (14)$$

3.2.2. Motion-Aware Local Kernel Smoothing

The temporally predicted EDF $\bar{\lambda}_{\pm}(t)$ may exhibit structural inconsistencies due to noise or threshold mismatch. To enforce local consistency, we introduce a **spatial EDF model** that performs motion-aware smoothing by incorporating neighborhood information. Unlike standard isotropic filters, spatial EDF patterns are directional and sparse, typically corresponding to moving edge intensities. Our model exploits this structure by imposing a directional motion prior on a 3×3 pixel neighborhood and expressing the smoothed EDF as a linear combination of directional basis patterns, whose coefficients are optimized via reconstruction error minimization.

We select 4 directional motion bases as

$$B_1(t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & b_1(t) & 0 \\ 0 & 0 & 1 \end{bmatrix} B_2(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & b_2(t) & 0 \\ 0 & 1 & 0 \end{bmatrix}$$
$$B_3(t) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & b_3(t) & 0 \\ 1 & 0 & 0 \end{bmatrix} B_4(t) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & b_4(t) & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad (15)$$

representing 0°, 45°, 90°, and 135° motion patterns, with $0 \le b_i(t) \le 1$ denoting the contribution to the center pixel. Then the smoothed EDF around pixel (x, y) is expressed

$$\Lambda_{\pm}(t) = \sum_{i=1}^{4} k_{\pm,i}(t) B_i(t), \quad (16)$$

where the optimal coefficients $\{k^*_{\pm,i}(t)\}_{i=1}^4$ are obtained by minimizing the q-norm reconstruction error:

$$\{k^*_{\pm,i}(t)\}_{i=1}^4 = \underset{\{k_{\pm,i}(t)\}_{i=1}^4}{\arg\min} \|\Lambda_{\pm}(t) - \bar{\Lambda}_{\pm}(t)\|_q, \quad (17)$$

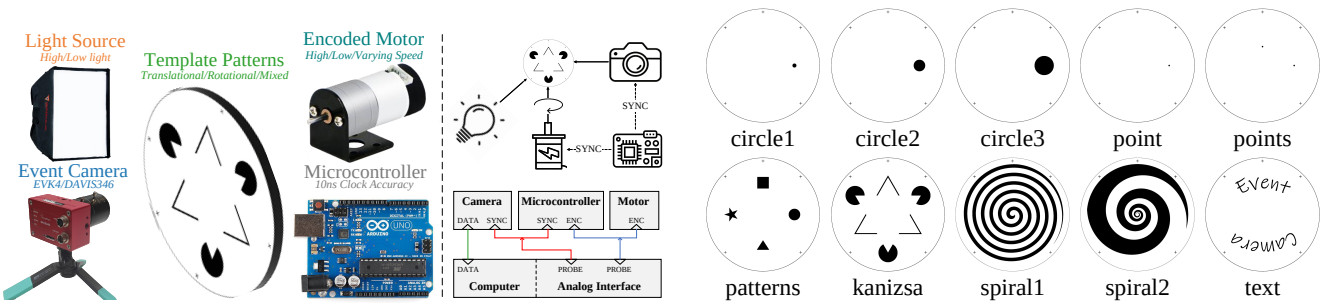
where $\bar{\Lambda}_{\pm}(t) = \left(\bar{\lambda}_{\pm}(t, x+i, y+j)\right)_{-1\le i,j\le 1}$ and $1 \le q \le 2$. We adaptively set $b_i(t) = \frac{1}{K(t)}$ if either neighboring side of the predicted EDF in the i-th direction is nonzero, otherwise $b_i(t) = 0$, with $K(t)$ being the number of effective bases. The updated central density flow is given by:

$$\lambda^*_{\pm}(t, x, y) = \sum_{i=1}^{4} k_{\pm,i}(t) b_i(t). \quad (18)$$

3.2.3. Density Flow Reconstruction and Event Sampling

Being able to evaluate $\lambda^*_{\pm}(t, x, y)$, the next thing to do is to reconstruct the continuous EDF. We employ a hybrid sampling scheme combining event-triggered asynchronous local sampling and periodic global sampling, preserving transient dynamics without abnormal integral drift.

An event triggers an update of the local 3×3 EDF neighborhood $\left(\lambda^*_{\pm}(t, x+i, y+j)\right)_{-1\le i,j\le 1}$, which depends on predicted EDF values from a 5×5 region



(a) **RED collection details.** A variety of synchronized intensity frames and events are generated by varying each component. Synchronization is achieved by attaching analog interface to probe the synchronization and encoder signal.

(b) **Printed binary patterns.** In addition to the rotational motion types, the *spiral1* and *spiral2* patterns additionally simulate linear motion types. The *point* and *points* patterns are only used for evaluating point tracking.

Figure 4. System setup of the Rotary Event Dataset benchmark.

$\left(\bar{\lambda}_{\pm}(t, x+i, y+j)\right)_{-2\le i,j\le 2}$. Additionally, a global update for all pixels is performed every T_s seconds, regardless of event activity. A zero-order interpolator is then used to obtain a continuous EDF representation, enabling event resampling by treating the process as a piecewise-homogeneous Poisson process.

3.2.4. An $O(1)$ Recursive Solver

All components of EDFilter are designed for asynchronous real-time operation with constant-time complexity $O(1)$.

Temporal EDF Prediction. From Eq. (11), if events are viewed as a sequence of polarity-modulated spike train

$$\mathcal{E}(t) = \sum_i p_i \delta(t - t_i), \tag{19}$$

we have $\psi(t) = \mathcal{E}(t) * \phi(t)$ with $*$ being the convolution. As this is a linear time-invariant (LTI) system, it admits the state-space representation

$$\frac{d\psi(t)}{dt} = \alpha(\mathcal{E}(t) - \mathcal{E}(t-h)). \tag{20}$$

Fixing h, $\lambda_{\pm}(t)$ and its integral follow from Eq. (10). The parameters α and $\beta_{\pm}$ are optimized independently, while h is chosen via a lookup table with discrete candidates maximizing likelihood, ensuring an $O(1)$ implementation.

Spatial EDF Update. Given the proven uniqueness of Eq. (18), solving Eq. (17) is a convex optimization problem with fixed-time convergence [2]. For $q = 2$, the solution simplifies to a 3×3 convolution; for $q = 1$, a local linear program provides the exact sparse solution. Both cases achieve constant-time complexity.

Event Sampling. Since EDF is modeled as piecewise-constant function, event intervals follow an exponential distribution and can be sampled sequentially. To handle discontinuities, a modified thinning algorithm of [27] incorporating event polarity is employed, also achieving $O(1)$ efficiency. Additional implementation details are provided in the supplementary material.

3.3. The Rotary Event Dataset Benchmark

Evaluating individual event quality is challenging because event counts and locations vary across sensors, and conventional cameras cannot provide microsecond-level reference intensity. To address this, we build a controlled high-speed rotary system that produces microsecond-accurate ground-truth intensity frames illustrated in Fig. 4(a). The setup includes a precision motor, a binary-patterned rotating disk, uniform illumination, an event camera, and a microcontroller for sub-microsecond synchronization. After spatial alignment using printed crosses and temporal alignment via shared-clock trigger pulses, we obtain tightly synchronized events and intensity frames.

Ten binary patterns as shown in Fig. 4(b) are printed on the disk to generate diverse motion types with uniform contrast. Spiral patterns create radial motion, while others produce rotational or mixed motions. Known pattern dimensions and the encoder's orientation readings allow reconstruction of ground-truth intensity at any timestamp. The motor uses a giant-magnetoresistance encoder with 120,000 steps per revolution (0.003°), calibrated using laser and photodiode measurements. Motor speed is PWM-controlled from 0 to 150 rpm. Two representative event cameras are used: (a) EVK4 [11] with 1280×720 spatial resolution and 1 μs temporal resolution and, (b) DAVIS346 [29] with 346×260 spatial resolution and 1 μs temporal resolution. To account for illumination effects, sequences are recorded under two lighting conditions and three rotation speeds, producing 120 highly synchronized sequences spanning about 20min. Point-based patterns are included for evaluating direct point tracking.

4. Experiments

In this section, we demonstrate the effectiveness of EDFilter in preserving both scene irradiance and motion information on various tasks and downstream applications.

Methods. We compare EDFilter with **Raw**, **EvFlow**

[31], **Ynoise** [10], **MLPF** [16], and **EventZoom** [6]. Here, **Raw** denotes the baseline that returns the unfiltered event stream. For EDFilter, we set the parameters to $t - s = 100$ms, $\gamma = 4$ms, LUT keys ranging from 156.25μs to 40ms with a $2\times$ increment, $q = 2$, and $T_s = 10$ms for event denoising; for direct point tracking, we set $p = 1$ and $T_s = \infty$ (fully asynchronous sampling). Additional details are provided in the supplementary material.

Datasets and Metrics. Besides the proposed RED dataset, we also evaluate on the E-MLB dataset [4], which contains extensive multi-level real-world noisy event sequences and provides the non-reference Event Structural Ratio (ESR) as an event quality metric.

For event denoising, we use mean ESR for evaluation on the E-MLB dataset and proposed to use normalized mean-squared-error (NMSE) on the RED dataset:

$$\text{NMSE} = \frac{\sum_{p\in\{-1,1\}} \sum_{t,x,y} (g_{t,x,y,p} - d_{t,x,y,p})^2}{\sum_{p\in\{-1,1\}} \sum_{t,x,y} g_{t,x,y,p}^2}, \quad (21)$$

where $g_{t,x,y,p}$ is the ground-truth radiance change and $d_{t,x,y,p}$ is the corresponding event density with polarity p at (t, x, y), where we use a temporal binning of 1ms for evaluation. For the direct point tracking task, we propose to use the average tracking error (ATE) for comparison:

$$\text{ATE} = \sum_i \frac{t_i - t_{i-1}}{T} \| P(x_i, y_i) - (c_x(t_i), c_y(t_i)) \|_2, \quad (22)$$

where (t_i, x_i, y_i) is the coordinate of the i-th event and $(c_x(t_i), c_y(t_i))$ is the coordinate of the closest point center at time t_i, with T being the time span of the evaluation period and $P(\cdot)$ being the projection operator from the image plane to the patterns plane.

4.1. Event Denoising

NMSE results on the RED dataset (Tab. 1) show that our method achieves the best overall performance. This indicates that the proposed filter preserves radiance-change information more accurately than competing approaches. Mean ESR results on the E-MLB dataset (Tab. 2) further confirm that our method best retains structural event information across diverse noise levels, consistent with the robustness reflected in the **std** column.

Qualitative results in Fig. 5 support these findings. EvFlow is overly aggressive and removes many real events. MLPF and Ynoise retain structure better, with MLPF generally performing second-best due to its richer local input representation and MLP discriminator. EventZoom, though visually plausible, performs poorly in NMSE because its fixed discretization sacrifices temporal precision and its convolutional modules diffuse spatial energy, degrading radiance-change accuracy. In contrast, our method adapts its temporal kernel to scene dynamics and enforces spatial sparsity via motion-aware local modeling and a q-norm reconstruction criterion, yielding cleaner and more faithful events.

4.2. Direct Point Tracking

The direct point tracking results on ATE (Tab. 3) show that our method ranks among the top two across all sequences. The closest competitor is EvFlow, which exhibits the lowest variance and thus the highest tracking stability. However, as evidenced by the filtering results in Fig. 5, EvFlow performs poorly in preserving accurate irradiance changes. The 3D visualizations in Fig. 6 further reveal that EvFlow retains only the most salient events along motion trajectories—beneficial for coarse motion tracking but detrimental to faithful radiance reconstruction. This highlights an intrinsic tension between accurate irradiance perception and motion perception, reflecting the separation of state and process information in conventional models.

By contrast, our method performs strongly on both fronts. This is achieved by modeling the complete event distribution through probability flux and optimizing it within a unified framework, thereby bridging the gap between state-information modeling and process-information modeling.

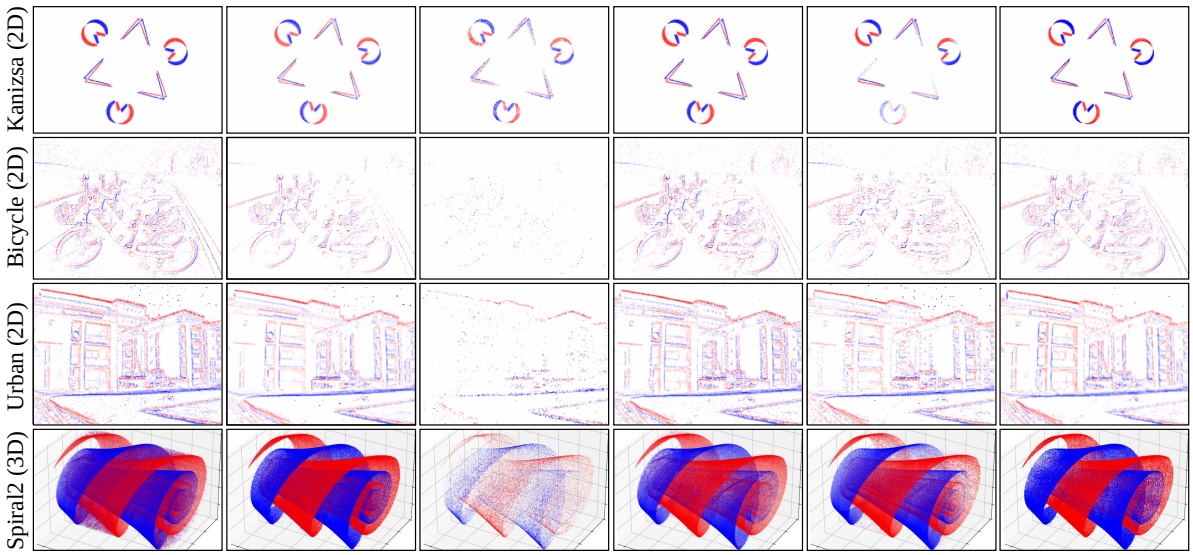
4.3. Analysis

Ablation Studies. Our pipeline consists of three modules. To assess their impact, we vary one component while keeping the others fixed. The averaged results for both filtering and direct point tracking are shown in Fig. 7. Substituting the temporal model shows that the proposed EDF formulation yields the best performance on both tasks. This advantage is particularly clear in motion perception, where Poisson [15] and Gaussian [21] temporal models rely on fixed correlation windows, whereas EDF adapts to scene dynamics through likelihood maximization. For spatial modeling, L1 better preserves motion accuracy, while L2 more faithfully maintains irradiance changes. Regarding the sampling strategy, asynchronous sampling gives the lowest ATE but higher NMSE than hybrid sampling (Async vs. Hybrid10ms). As discussed in Sec. 4.2, a more aggressive sampling is preferable when motion is the primary objective. Synchronous sampling performs worst across all metrics, underscoring its inefficiency for inherently asynchronous event signals. These observations also indicate that extremely fine sampling (Sync1ms vs. Sync10ms) offers limited gains relative to its computational cost.

Runtime. We evaluate the runtime characteristics of our method for online deployment. Since event streams are asynchronous, two complementary metrics are used: latency, measuring the time to produce an output after a single event arrives (online test), and throughput, measuring the sustained event-processing rate over time (offline/online test). We benchmark three representative methods—MLPF, EventZoom, and our EDFilter—under identical conditions

Table 1. NMSE results of RED. DVS/EVK columns: DAVIS346 and EVK4 sequences. Bold red/blue: 1st/2nd best values.

Method	circle1		circle2		circle3		kanizsa		patterns		spiral1		spiral2		text		std	
	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK	DVS	EVK
Raw	0.106	0.390	**0.087**	0.166	**0.060**	0.118	**0.061**	0.054	**0.085**	0.087	**0.137**	0.139	**0.094**	0.139	0.058	0.052	**0.045**	0.140
EventZoom [6]	0.223	0.338	0.111	0.109	0.084	0.076	0.210	0.206	0.129	0.129	0.331	0.108	0.120	0.108	0.283	0.227	0.168	0.130
EvFlow [31]	0.285	0.504	0.265	0.378	0.242	0.361	0.094	0.212	0.144	0.214	0.529	0.312	0.346	0.312	0.065	0.142	0.198	0.163
MLPF [16]	**0.104**	**0.226**	0.091	**0.073**	0.066	**0.054**	**0.062**	**0.049**	0.089	**0.072**	0.250	**0.094**	0.134	**0.094**	**0.057**	**0.049**	0.090	**0.057**
Ynoise [10]	0.244	0.230	0.186	0.125	0.159	0.072	0.162	0.082	0.173	0.101	0.516	0.115	0.309	0.115	0.187	0.089	0.160	0.068
Ours	**0.096**	**0.193**	**0.075**	**0.059**	**0.055**	**0.044**	0.071	**0.043**	**0.079**	**0.062**	**0.121**	**0.086**	**0.088**	**0.086**	**0.058**	**0.038**	**0.036**	**0.051**



(a) Raw (b) EventZoom [6] (c) EvFlow [31] (d) MLPF [16] (e) Ynoise [10] (f) Ours

Figure 5. Visualization of filtered events as 2D event frames (top 3 rows) and 3D point clouds (bottom row). Sequences come from the proposed RED dataset (rows 1,4), the E-MLB dataset [4] (row 2) and the ECD dataset [26] (row 3).

Table 2. **Mean ESR results on E-MLB**. ND%: noise level. Bold red/blue: 1st/2nd best values.

Method	E-MLB (Day/Night)			
	ND1	ND4	ND16	ND64
Raw	0.82/0.89	0.82/0.82	0.82/0.79	0.79/0.77
EventZoom [6]	**1.00**/**1.06**	**0.99**/**1.01**	**1.00**/**1.01**	**0.97**/**0.99**
EvFlow [31]	0.85/0.97	0.88/0.98	0.87/0.89	0.83/0.80
MLPF [16]	0.85/0.93	0.89/0.93	0.85/0.91	0.84/0.91
Ynoise [10]	0.87/1.01	0.86/0.94	0.86/0.88	0.82/0.79
Ours	**1.02**/**1.07**	**1.02**/**1.03**	**1.02**/**1.00**	**1.00**/**0.99**

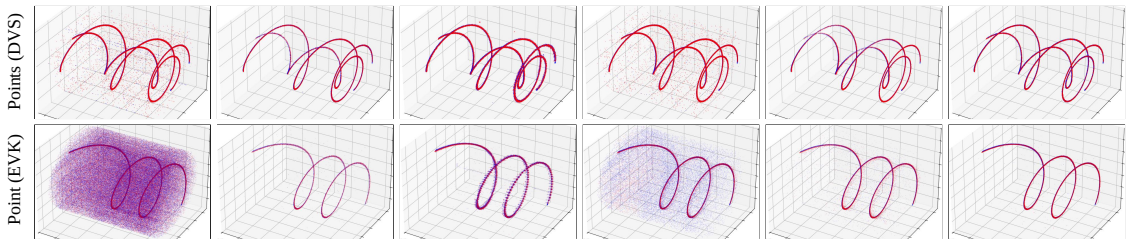
Table 3. **ATE results in millimeters (mm).** DVS/EVK columns: DAVIS346 and EVK4 sequences. Red/blue: 1st/2nd best values.

Method	point		points		std	
	DVS	EVK	DVS	EVK	DVS	EVK
Raw	7.542	46.319	4.711	30.728	1.787	24.665
EventZoom [6]	4.168	4.711	3.680	3.084	2.096	2.672
EvFlow [31]	**2.290**	**2.978**	**2.306**	**2.125**	**0.581**	**1.071**
MLPF [16]	7.005	28.811	4.499	17.444	1.540	11.635
Ynoise [10]	2.805	7.223	3.428	4.444	1.031	7.864
Ours	**2.358**	**2.954**	**2.518**	**2.046**	**0.810**	**1.297**

with 346×260 resolution on a single core of an R9-7945HX CPU, as shown in Tab. 4. For synchronous EventZoom, throughput is computed on real events with an average rate of 1.23 events/μs. Our method achieves the lowest latency at 4.99 μs, outperforming MLPF by roughly 51%, demonstrating that physically grounded signal processing can still operate at microsecond scale. EventZoom has much higher latency due to the event-frame accumulation and heavy frame-processing blocks. In terms of throughput, EDFilter can already process 210k events/s surpassing EventZoom.

4.4. Applications

Event-Based SLAM. The reference SLAM algorithm is selected as ESVO [36], which uses stereo semi-global matching to recover camera trajectories. The performance is evaluated by comparing the computed trajectories and



(a) Raw (b) EventZoom [6] (c) EvFlow [31] (d) MLPF [16] (e) Ynoise [10] (f) Ours

Figure 6. Visualization of filtered events in 3D as point clouds. Sequences are captured with DAVIS346 (top row) and EVK4 (bottom row).

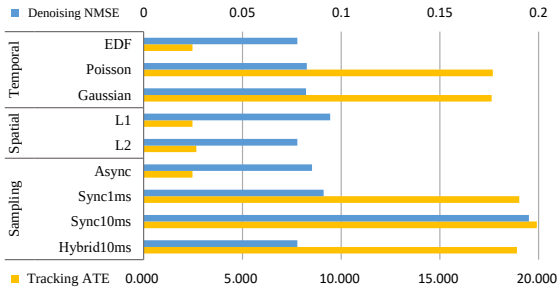


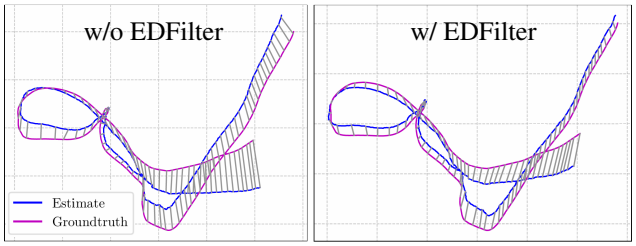
Figure 7. Ablation studies of EDFilter by its temporal modeling, spatial modeling, and sampling paradigm.

Table 4. **Latency and throughput comparison of event-signal filters.** Latency is measured in seconds (s), and throughput is measured in events per second (events/s).

Metric	MLPF [16]	EventZoom [6]	Ours
Latency	7.55e-6	1.21e0	4.99e-6
Throughput	4.73e6	1.34e5	2.10e5

ground-truth trajectories on the MVSEC [37] and RPG [35] datasets. The absolute translational error (RMS) results and the top view of computed trajectories are provided in Fig. 8. After inserting the proposed EDFilter, the tracking performance is improved for most sequences, especially for those previously having poor results like *indoor_flying1* and *indoor_flying3*. Such results are a direct demonstration of the motion preservation ability of the proposed method.

Video Reconstruction. The reference video reconstruction algorithm is selected as E2VID [28], which learns to generate corresponding video frames from input event frames with a recurrent network. As with evaluating other data preprocessing pipelines, we fine-tune the pretrained model on filtered events to obtain the final reconstruction results. The performance is assessed on the E-MLB [4] dataset, reporting the average peak-to-signal-ration (PSNR), mean-squared error (MSE), structural similarity (SSIM) index, and perceptual similarity (LPIPS) [34] and reconstructed frames in Fig. 9. The reconstruction results have been improved for all the metrics, which is also evident visually. Such results clearly demonstrate the benefit of EDFilter in irradiance preservation.



Method	indoor_flying1	indoor_flying3	rpg_bin	rpg_boxes	rpg_desk	rpg_monitor
w/o EDFilter	18.4	10.1	3.0	6.1	**3.5**	2.3
w/ EDFilter	**15.5**	**8.8**	**2.9**	**4.8**	3.9	**2.2**

Figure 8. SLAM results on the MVSEC [37] and RPG [35] datasets. Top: top-view trajectory of *indoor_flying1*. Bottom: RMS tracking error in centimeters (cm). Bold red: best value.



Method	PSNR↑	MSE↓	SSIM↑	LPIPS↓
w/o EDFilter	16.04	0.035	0.53	0.30
w/ EDFilter	**18.37**	**0.019**	**0.68**	**0.24**

Figure 9. Video reconstruction results on the E-MLB [4] dataset. Top: reconstructed frame. Bottom: metrics. Bold red: best value.

5. Conclusion

We proposed EDFilter, an event signal filtering framework grounded in a probability-flux interpretation of event generation. By estimating the threshold-crossing probability flux through efficient temporal KDE, motion-aware spatial smoothing, and asynchronous resampling, EDFilter provides physically consistent and real-time event reconstruction. The RED dataset with microsecond irradiance ground truth is also captured for evaluation. Experiments show that EDFilter achieves superior denoising accuracy and tracking performance, highlighting probability-flux modeling as a promising direction for future research on generative models, physics-informed neural representations, and probabilistic formulations in event-based vision.

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References

[1] R Baldwin, Mohammed Almatrafi, Vijayan Asari, and Keigo Hirakawa. Event probability mask (epm) and event denoising convolutional neural network (edncnn) for neuromorphic cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1701–1710, 2020. 2

[2] Stephen P Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge university press, 2004. 5

[3] Tobi Delbruck et al. Frame-free dynamic digital vision. In *Proceedings of Intl. Symp. on Secure-Life Electronics, Advanced Electronics for Quality Life and Society*, pages 21–26. Citeseer, 2008. 2

[4] Saizhe Ding, Jinze Chen, Yang Wang, Yu Kang, Weiguo Song, Jie Cheng, and Yang Cao. E-mlb: Multilevel benchmark for event-based camera denoising. *IEEE Transactions on Multimedia*, 2023. 1, 6, 7, 8, 9

[5] Peiqi Duan, Zihao W Wang, Boxin Shi, Oliver Cossairt, Tiejun Huang, and Aggelos K Katsaggelos. Guided event filtering: Synergy between intensity images and neuromorphic events for high performance imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11): 8261–8275, 2021. 2

[6] Peiqi Duan, Zihao W Wang, Xinyu Zhou, Yi Ma, and Boxin Shi. Eventzoom: Learning to denoise and super resolve neuromorphic events. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12824–12833, 2021. 2, 6, 7, 8, 10

[7] Peiqi Duan, Yi Ma, Xinyu Zhou, Xinyu Shi, Zihao W Wang, Tiejun Huang, and Boxin Shi. Neurozoom: Denoising and super resolving neuromorphic events and spikes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023. 2

[8] JN Elgin. The fokker-planck equation: methods of solution and applications. *Optica Acta: International Journal of Optics*, 31(11):1206–1207, 1984. 3, 2

[9] Davide Falanga, Kevin Kleber, and Davide Scaramuzza. Dynamic obstacle avoidance for quadrotors with event cameras. *Science Robotics*, 5(40):eaaz9712, 2020. 1

[10] Yang Feng, Hengyi Lv, Hailong Liu, Yisa Zhang, Yuyao Xiao, and Chengshan Han. Event density based denoising method for dynamic vision sensor. *Applied Sciences*, 10(6): 2024, 2020. 2, 6, 7, 8

[11] Thomas Finateu, Atsumi Niwa, Daniel Matolin, Koya Tsuchimoto, Andrea Mascheroni, Etienne Reynaud, Pooria Mostafalu, Frederick Brady, Ludovic Chotard, Florian LeGoff, et al. 5.10 a 1280× 720 back-illuminated stacked temporal contrast event-based vision sensor with 4.86 μm pixels, 1.066 geps readout, programmable event-rate controller and compressive data-formatting pipeline. In *2020 IEEE International Solid-State Circuits Conference-(ISSCC)*, pages 112–114. IEEE, 2020. 1, 5, 7

[12] Maxim Finkelstein. *Failure rate modelling for reliability and risk*. Springer, 2008. 3

[13] Guillermo Gallego, Henri Rebecq, and Davide Scaramuzza. A unifying contrast maximization framework for event cameras, with applications to motion, depth, and optical flow estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3867–3876, 2018. 1

[14] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J Davison, Jörg Conradt, Kostas Daniilidis, et al. Event-based vision: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(1):154–180, 2020. 1

[15] Cheng Gu, Erik Learned-Miller, Daniel Sheldon, Guillermo Gallego, and Pia Bideau. The spatio-temporal poisson point process: A simple model for the alignment of event camera data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 13495–13504, 2021. 6

[16] Shasha Guo and Tobi Delbruck. Low cost and latency event camera background activity denoising. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(1):785–795, 2022. 2, 6, 7, 8

[17] Shasha Guo, Chenyang Shi, Lei Wang, Jing Jin, and Yuliang Lu. Ebf: An event-based bilateral filter for effective neuromorphic vision sensor denoising. *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. 2

[18] Yuhuang Hu, Shih-Chii Liu, and Tobi Delbruck. v2e: From video frames to realistic dvs events. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1312–1321, 2021. 1

[19] Bin Jiang, Bo Xiong, Bohan Qu, M Salman Asif, You Zhou, and Zhan Ma. Edformer: Transformer-based event denoising across varied noise levels. In *European Conference on Computer Vision*, pages 200–216. Springer, 2024. 2

[20] Alireza Khodamoradi and Ryan Kastner. $o(n)$ o (n)-space spatiotemporal filter for reducing noise in neuromorphic vision sensors. *IEEE Transactions on Emerging Topics in Computing*, 9(1):15–23, 2018. 2

[21] Jianing Li, Yihua Fu, Siwei Dong, Zhaofei Yu, Tiejun Huang, and Yonghong Tian. Asynchronous spatiotemporal spike metric for event cameras. *IEEE Transactions on Neural Networks and Learning Systems*, 2021. 6

[22] Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck. A 128 × 128 120 db 15μs latency asynchronous temporal contrast vision sensor. *IEEE journal of solid-state circuits*, 43 (2):566–576, 2008. 1

[23] Songnan Lin, Ye Ma, Zhenhua Guo, and Bihan Wen. Dvs-voltmeter: Stochastic process-based event simulator for dynamic vision sensors. In *European Conference on Computer Vision*, pages 578–593. Springer, 2022. 2

[24] Hongjie Liu, Christian Brandli, Chenghan Li, Shih-Chii Liu, and Tobi Delbruck. Design of a spatiotemporal correlation

filter for event-based sensors. In *2015 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 722–725. IEEE, 2015. 2

[25] Elias Mueggler, Christian Forster, Nathan Baumli, Guillermo Gallego, and Davide Scaramuzza. Lifetime estimation of events from dynamic vision sensors. In *2015 IEEE international conference on Robotics and Automation (ICRA)*, pages 4874–4881. IEEE, 2015. 2

[26] Elias Mueggler, Henri Rebecq, Guillermo Gallego, Tobi Delbruck, and Davide Scaramuzza. The event-camera dataset and simulator: Event-based data for pose estimation, visual odometry, and slam. *The International journal of robotics research*, 36(2):142–149, 2017. 7

[27] Yosihiko Ogata. On lewis' simulation method for point processes. *IEEE transactions on information theory*, 27(1):23–31, 1981. 5

[28] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. High speed and high dynamic range video with an event camera. *IEEE transactions on pattern analysis and machine intelligence*, 43(6):1964–1980, 2019. 8

[29] Gemma Taverni, Diederik Paul Moeys, Chenghan Li, Celso Cavaco, Vasyl Motsnyi, David San Segundo Bello, and Tobi Delbruck. Front and back illuminated dynamic and active pixel vision sensors comparison. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 65(5):677–681, 2018. 1, 5, 7

[30] Antoni Rosinol Vidal, Henri Rebecq, Timo Horstschaefer, and Davide Scaramuzza. Ultimate slam? combining events, images, and imu for robust visual slam in hdr and high-speed scenarios. *IEEE Robotics and Automation Letters*, 3(2):994–1001, 2018. 1

[31] Yanxiang Wang, Bowen Du, Yiran Shen, Kai Wu, Guangrong Zhao, Jianguo Sun, and Hongkai Wen. Ev-gait: Event-based robust gait recognition using dynamic vision sensors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6358–6367, 2019. 2, 6, 7, 8

[32] Zihao W Wang, Peiqi Duan, Oliver Cossairt, Aggelos Katsaggelos, Tiejun Huang, and Boxin Shi. Joint filtering of intensity images and neuromorphic events for high-resolution noise-robust imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1609–1619, 2020. 2

[33] Pei Zhang, Zhou Ge, Li Song, and Edmund Y Lam. Neuromorphic imaging with density-based spatiotemporal denoising. *IEEE Transactions on Computational Imaging*, 2023. 2

[34] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. 8

[35] Yi Zhou, Guillermo Gallego, Henri Rebecq, Laurent Kneip, Hongdong Li, and Davide Scaramuzza. Semi-dense 3d reconstruction with a stereo event camera. In *Proceedings of the European conference on computer vision (ECCV)*, pages 235–251, 2018. 8

[36] Yi Zhou, Guillermo Gallego, and Shaojie Shen. Event-based stereo visual odometry. *IEEE Transactions on Robotics*, 37 (5):1433–1450, 2021. 7

[37] Alex Zihao Zhu, Dinesh Thakur, Tolga Özaslan, Bernd Pfrommer, Vijay Kumar, and Kostas Daniilidis. The multivehicle stereo event camera dataset: An event camera dataset for 3d perception. *IEEE Robotics and Automation Letters*, 3 (3):2032–2039, 2018. 8